

Tim Jacobs

Staff Software Engineer | Data, AI & Platform Architecture

CO 80501, Longmont, Colorado (US)

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[in](#) Tim Jacobs

SUMMARY

Staff Software Engineer with deep expertise in large-scale data platforms, LLM-powered retrieval systems (RAG), and cloud-native infrastructure. Proven architect of event-driven pipelines, vector search applications, and shared platform components that power data-intensive products. 15+ years building high-leverage backend systems from storage validation to distributed AI services. Seeking to apply this experience to mission-critical domains including healthcare, climate, education, or civic infrastructure, where technical excellence drives positive human impact.

SKILLS

Data Engineering, AI & Platform Architecture

Expert

Snowflake dbt Apache Airflow

ETL/ELT pipeline design

Vector search & embeddings

RAG architecture

Event-driven systems

Data warehouse design

AI & Machine Learning

Accomplished

LLM integration & prompting

Snowflake Cortex Streamlit

Custom tokenization & vectorization

k-NN / semantic search

Python ML ecosystem

Recommendation systems

Data quality automation

Cloud Architecture & DevOps

Expert

AWS Microservices architecture

Docker & containers

CI/CD pipeline design

Infrastructure-as-Code

Serverless architecture

Software Development

Expert

Python SQL Go/Golang

Database design & optimization

Event-driven microservices

Engineering Leadership

Expert

Technical architecture & roadmap

Team mentorship & growth

Agile/Scrum leadership

Cross-functional collaboration

DevOps transformation

WORK EXPERIENCE (7)

Dec 2022 - Current

Senior Software Engineer at [EDO Inc.](#)

Data platform & AI systems architecture for cross-platform content intelligence.

- Architected and deployed an internal RAG-based data-quality tool using Snowflake Cortex for LLM inference, custom tokenization and vectorization pipelines, and k-NN/embedding similarity search to surface AI-generated taxonomy mappings.
- Built a Streamlit review interface so the data quality team can inspect, validate, and accept AI recommendations produced by the RAG pipeline.

- Implemented medallion architecture on Snowflake-based data pipelines to ingest and process trillions of rows of cross-platform data, supporting 20+ enterprise partners.
- Architected a comprehensive data deletion and anonymization system in Snowflake and Airflow to ensure legal and regulatory compliance; established flexible automated TTL policies for sensitive PII data across large-scale distributed datasets.
- Engineered a hypernormalized metadata management system that standardized fragmented data from dozens of sources into a single canonical schema, achieving 80% coverage of all monthly events and eliminating manual mapping bottlenecks.

Oct 2020 - Dec 2022

Staff Software Engineer at [Pie Insurance](#)

Full-stack platform engineering, quality-focused technical leadership, and operational data architecture in a microservices environment.

- Architected and deployed 40+ shared platform components consumed by multiple engineering teams; reduced time-to-market for new features by 50% and improved code reuse across the organization.
- Led technical design of CI/CD pipeline modernization; improved deployment velocity by 3x, reduced mean time to recovery by 60%, and enabled 200+ deployments per week.
- Designed comprehensive test architecture for an event-driven billing service; enabled continuous deployment across 8+ microservices-based systems with 99.9% reliability.
- Led operational data team; designed and implemented reference data re-architecture using event-driven ETL; reduced mean commit-to-deploy time from weeks to hours, increasing release velocity by 3x.
- Developed scalable test automation framework for microservices in Python/Golang using AWS Lambda, S3, and SQS; automated 90% of previously manual test execution, reducing test cycle time from 6 hours to 12 minutes.
- Mentored 6 junior engineers on microservices patterns, testing strategy, and AWS cloud best practices; 3 engineers promoted during tenure.

Apr 2016 - Oct 2020

Senior Software Engineer at [NetApp - Solidfire](#)

Platform architecture, DevOps transformation, and test infrastructure leadership.

- Designed and led a test result reporting platform from concept to production; engineered a queryable database with 2M+ daily test records and automated management dashboards, eliminating 40+ hours per week of manual reporting.
- Spearheaded a DevOps transformation initiative with cross-functional stakeholders; reduced build failure rate by 50%, improved deployment reliability by 35%, and decreased infrastructure costs by \$500K+ per year.
- Architected a containerized test framework enabling migration of 1,000+ test cases from legacy systems to modern CI-driven infrastructure; reduced test execution time by 70% and improved reproducibility.

Jun 2013 - Apr 2016

SSD Validation Engineer → **Engineering Supervisor** at [Micron Technology](#)

Validation leadership, data-driven operations, and team management (4 direct reports).

- Managed a team of 4 engineers validating enterprise/data center SAS/SATA SSDs across 4 Tier-1 OEM partners; maintained >\$60M quarterly revenue through rigorous, on-time qualification processes.
- Designed and deployed a qualification test status database with automated test script integration; eliminated manual reporting, enabled real-time visibility into validation progress, and reduced management reporting overhead by 30 hours per week.
- Led a complete automation rewrite to meet compatibility team requirements; reduced deployment schedule by 20 months, improving test execution efficiency by orders of magnitude.

Apr 2011 - May 2013

Senior Test Engineer at Western Digital

Automated failure analysis and firmware debug tooling.

- Developed automated failure analysis debug scripts for the firmware team; reduced debug overhead by 30% and accelerated time-to-resolution by 2x, enabling faster release cycles.

May 2009 - Mar 2011

Field Engineer at Parsons Infrastructure & Technology

Field automation and systems integration.

- Built a comprehensive instrument, material, schedule, and design verification database with a web interface and automated work control document generation; eliminated field rework, reducing engineering overhead from 40+ hours per week to zero.

Dec 2005 - Apr 2009

Hardware Test Engineer → **Failure Analysis Lead** at Seagate Technology

Failure analysis leadership and production transfer.

- Led product transfer from development (Longmont, CO) to production (Wuxi, China) as failure analysis lead; delivered Seagate's first 7200 RPM mobile drive to production at >90% yields.

PROJECTS (1)

RAG Taxonomy Mapping Review Tool

Dec 2022 - Current

RAG Snowflake Cortex Streamlit k-NN search Tokenization Vectorization Data quality

Internal data-quality application that uses retrieval-augmented generation to recommend taxonomy mappings for cross-platform content catalogs.

- Developed a custom tokenizer and vectorizer to prepare content metadata for k-NN similarity search and LLM inference inside Snowflake Cortex.
- Implemented the RAG pipeline end-to-end, from vectorization and retrieval to LLM-based ranking of candidate taxonomy mappings.
- Built a Streamlit interface so the data quality team can review, verify, and accept AI-generated recommendations before they enter production data pipelines.
- Supports large-scale, cross-platform content catalogs and reduces manual taxonomy mapping overhead.

EDUCATION (1)

- 2005

Bachelor of Science Electrical Engineering at University of Colorado at Boulder

Neural Network Design Bioengineering focus

Grade: 3.74 (Cum Laude)

LANGUAGES

English

Native speaker

INTERESTS

Machine Learning & AI
Research

- Neural network design
- LLM applications & RAG systems
- Predictive analytics

Civic & Open
Technology

- Open-source contributions
- Technology for social good

Outdoor Recreation

- Hiking
- Trail running
- Mountain biking

REFERENCES

Available upon request